

Worksheet 2: The Qubit and Quantum Mechanics Fundamentals

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The six cardinal states.

$$|0\rangle, |1\rangle, |\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle), \quad |\pm i\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm i|1\rangle).$$

The Born rule. Measuring $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ in the computational basis returns 0 with probability $|\alpha|^2$ and 1 with probability $|\beta|^2$, and leaves the qubit in the corresponding state. Normalisation requires $|\alpha|^2 + |\beta|^2 = 1$. For a measurement in any other basis $\{|u\rangle, |u^\perp\rangle\}$, the probability of the outcome u is $|\langle u|\psi\rangle|^2$.

Constants. $h = 6.626 \times 10^{-34} \text{ J s}$, $k_B = 1.381 \times 10^{-23} \text{ J/K}$, $\hbar\omega = hf$.

Part A — States, phases and measurement

1. One state, three measurements. Consider

$$|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4i}{5}|1\rangle.$$

- Check that $|\psi\rangle$ is normalised, and give the probabilities of the two outcomes of a Z (computational-basis) measurement.
- Compute the probabilities of the two outcomes of an X measurement, i.e. $|\langle +|\psi\rangle|^2$ and $|\langle -|\psi\rangle|^2$.
- Do the same for a Y measurement, using $|\pm i\rangle$. Then say which of the three measurements is almost perfectly predictable on this state and which is a fair coin, and state roughly where $|\psi\rangle$ sits on the Bloch sphere.

2. Which phases can be seen.

- Let $|\phi\rangle = e^{i\gamma}|\psi\rangle$ for a real constant γ . Show that for *any* measurement basis $\{|u\rangle, |u^\perp\rangle\}$ the two states give identical probabilities. Conclude that a global phase is not physical.
- Now take the family $|\psi_\varphi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\varphi}|1\rangle)$. Show that the Z statistics do not depend on φ at all, but that

$$p(+) = |\langle +|\psi_\varphi\rangle|^2 = \frac{1}{2}(1 + \cos \varphi).$$

- In one sentence, why is the relative phase — and not the global phase — described as the thing every quantum algorithm spends?

3. The sequential Stern–Gerlach experiment. A beam of silver atoms passes through a magnet aligned along z , which separates it into two beams; the lower beam ($|1\rangle$) is blocked. The surviving beam then passes through a magnet aligned along x , and only the $|+\rangle$ output is kept. That beam finally enters a third magnet aligned along z .

- What is the state of every atom leaving the first magnet?
- What fraction of those atoms survives the second magnet, and in what state do they leave it?
- What does the third magnet do? Give the fraction of the original surviving beam that ends up in each of its two outputs.
- Suppose each atom secretly carried a definite pair of values, one for z and one for x , fixed before it entered the apparatus. What would that model predict for the third magnet, and how does the experiment contradict it?

Part B — What a qubit is made of

4. Choosing two levels, and keeping them cold. A qubit is a chosen pair of levels inside a larger spectrum. Two conditions make the choice work: a large gap, $\hbar\omega_{01} \gg k_B T$, and an anharmonic ladder, $\omega_{12} \neq \omega_{01}$.

- (a) A superconducting qubit has $\omega_{01}/2\pi = 5$ GHz. Find the temperature at which $\hbar\omega_{01} = k_B T$.
- (b) Estimate the thermal population of the excited state at $T = 15$ mK using the Boltzmann factor $e^{-\hbar\omega_{01}/k_B T}$. Comment on why these machines sit at 10–20 mK rather than, say, at 1 K.
- (c) Explain why a perfect harmonic oscillator cannot serve as a qubit, and say what *leakage* is. Why is leakage a more awkward error than a simple bit flip?

5. Driving a qubit with a pulse. A resonant pulse rotates the Bloch vector about an axis in the equatorial plane at a rate given by the Rabi frequency Ω . Starting from $|0\rangle$,

$$p(1) = \sin^2\left(\frac{\Omega t}{2}\right), \quad \text{pulse area } \Theta = \Omega t.$$

- (a) A device has $\Omega/2\pi = 50$ MHz. How long is a π pulse, and which gate does it perform? How long is a $\pi/2$ pulse, and what does it do to $|0\rangle$?
- (b) Sketch $p(1)$ against Ωt from 0 to 2π . Why is sweeping the pulse duration the first measurement made on a newly built device?
- (c) If the drive frequency misses the qubit frequency by Δ , the rotation axis tilts out of the equatorial plane by $\arctan(\Delta/\Omega)$, the rate becomes $\sqrt{\Omega^2 + \Delta^2}$, and the maximum reachable excitation is $\Omega^2/(\Omega^2 + \Delta^2)$. Evaluate all three for $\Delta = \Omega$, and explain in one line why the qubit can no longer be fully inverted.

6. Losing the state.

- (a) Define T_1 and T_2 , and describe what each does to the Bloch vector. Why is $T_2 \leq 2T_1$ rather than the two being independent?
- (b) The number of gates available before the information is gone is roughly $N_{\text{gates}} \approx T_2/t_{\text{gate}}$. Evaluate it for a superconducting qubit with $T_2 = 100 \mu\text{s}$ and a two-qubit gate of 100 ns, and for a trapped ion with $T_2 = 10$ s and a two-qubit gate of 100 μs . Which platform is better, and what does the comparison say about quoting a coherence time on its own?
- (c) In what sense is dephasing the environment performing a measurement on your qubit? Relate your answer to what happens when a measurement is inserted between the two Hadamards of the interference circuit from Lecture 1.

Part C — Building real machines

7. Why you cannot copy a qubit. Suppose a gate C acts on a qubit in an unknown state together with a blank qubit, and copies the two basis states:

$$C|0\rangle|0\rangle = |0\rangle|0\rangle, \quad C|1\rangle|0\rangle = |1\rangle|1\rangle.$$

- (a) Gates are linear. Use this to compute $C|+\rangle|0\rangle$, where $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$.
- (b) Expand $|+\rangle|+\rangle$, the state a genuine copier would have to produce. Compare it with your answer to (a) and conclude.
- (c) Quantum error correction nevertheless works. Explain in two or three sentences how it evades the theorem — what does the circuit measure, if not the state?

8. Comparing platforms.

- (a) State DiVincenzo's five criteria for a physical system to serve as a quantum computer.
- (b) Superconducting circuits run at 10–20 mK with two-qubit gates of order 10^2 ns and fixed nearest-neighbour connectivity. Trapped ions run at room temperature with two-qubit gates of order 10 – 10^2 μ s and effectively all-to-all connectivity within a chain. For each platform, name the criterion on which it scores best and the one that gives it the most trouble.
- (c) A good two-qubit gate fails with probability of order 10^{-3} , while an algorithm running a billion gates needs an error rate below about 10^{-9} . Given that gap, explain in one or two sentences why error correction is not optional, and what is meant by a *threshold*.